PARTE 2

Expressões e operadores aritméticos

Variáveis, tipos, entrada/saída e expressões em C

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Data:	**03\09**	**Professor:**	Diego Abreu

Objetivos

- Construir expressões com variáveis, valores e operadores.
- Utilizar os operadores +, -, *, / e %.
- Compreender resto da divisão e divisão inteira.

Conceito - Expressões

Uma expressão é uma combinação de valores, variáveis e operadores que produz um resultado.

Exemplos

```
a + b
nota1 + nota2 + nota3
largura * altura
```

Teste rápido

```
int a = 5;
int b = 3;

printf("%d\n", a + b);
```

Operadores aritméticos

+	adição
-	subtração
*	multiplicação
/	divisão
%	resto da divisão inteira

Atividade 1 - Prevendo resultados

```
int a = 10;
int b = 3;

a + b
a - b
a * b
a / b
a % b
```

Antes de executar, escreva o resultado esperado de cada expressão. Depois, confirme no programa. **1. Resultado de a + b**

13_____

2. Resultado de a - b:

_____7_____

3. Resultado de a * b:

_____30_____

4. Resultado de a / b:

_____3_____

5. Resultado de a % b:

_____1_____

O operador módulo %

O módulo devolve o resto da divisão inteira. Em 10 ÷ 3, o quociente é 3 e o resto é 1; portanto, 10 % 3 produz 1.

```
8 % 2 -> 0
9 % 2 -> 1
17 % 5 -> 2
```

Aplicação futura: O módulo será útil em problemas como verificar paridade e decompor unidades de tempo.

Divisão inteira

```
int a = 10;
int b = 3;

printf("%d\n", a / b);
```

Resultado

```
3
```

Regra-chave: Quando os dois operandos são int, a divisão é inteira. A parte fracionária é descartada.

Atividade 2 - Soma e operações

Código-base

```
#include <stdio.h>

int main(void) {
 int numero1;
 int numero2;
 int soma;

 printf("Primeiro numero: ");
 scanf("%d", &numero1);

 printf("Segundo numero: ");
 scanf("%d", &numero2);

 soma = numero1 + numero2;

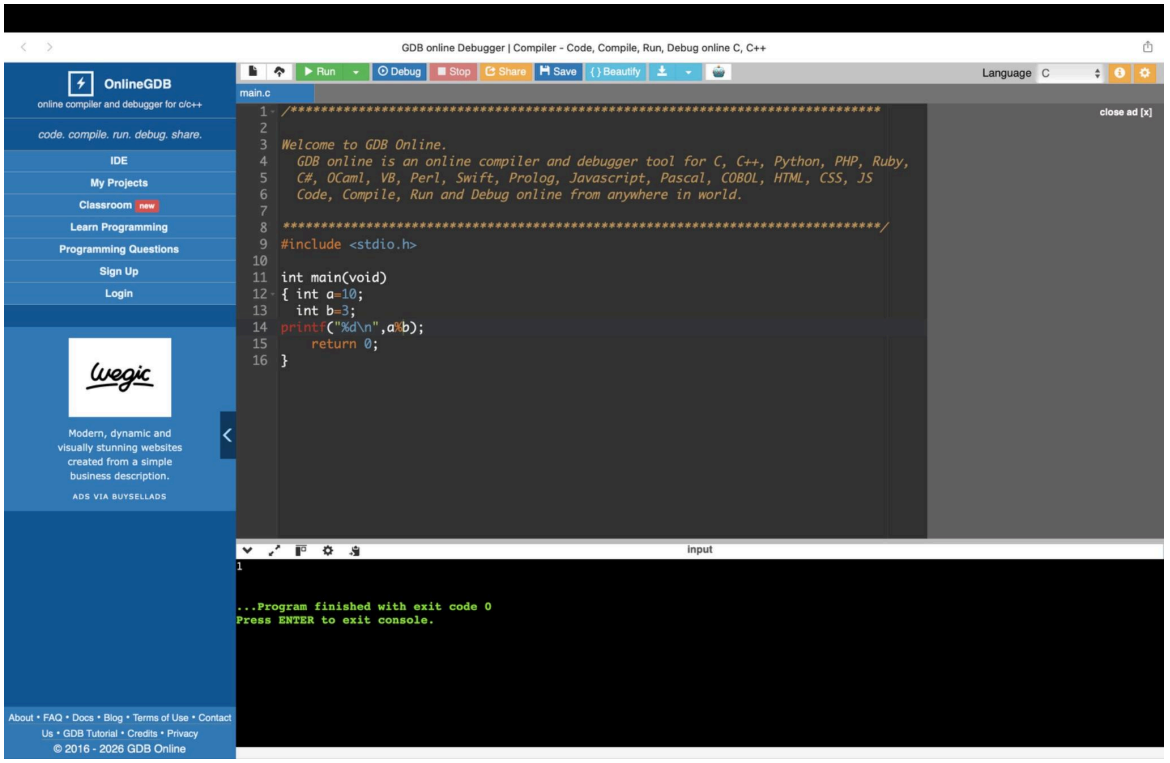
 printf("Soma = %d\n", soma);
 return 0;
}
```

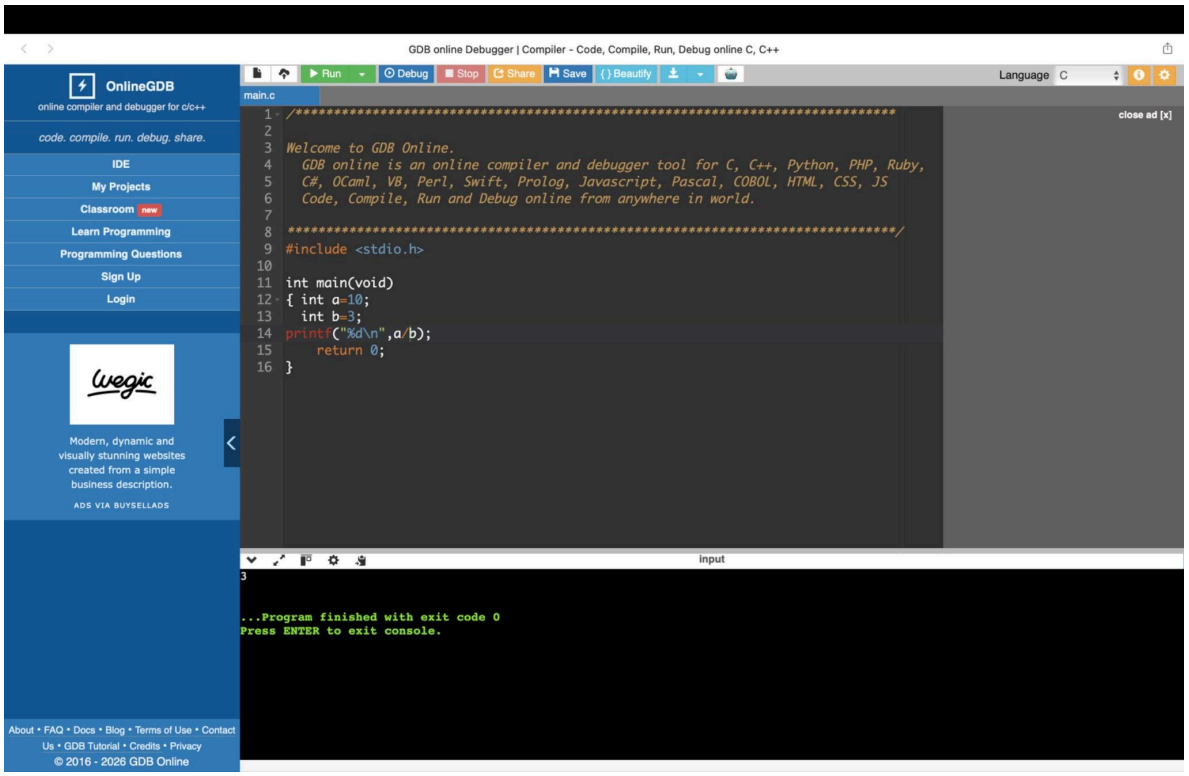
• Execute o programa.
• Acrescente subtração.
• Acrescente multiplicação.
• Acrescente divisão inteira.

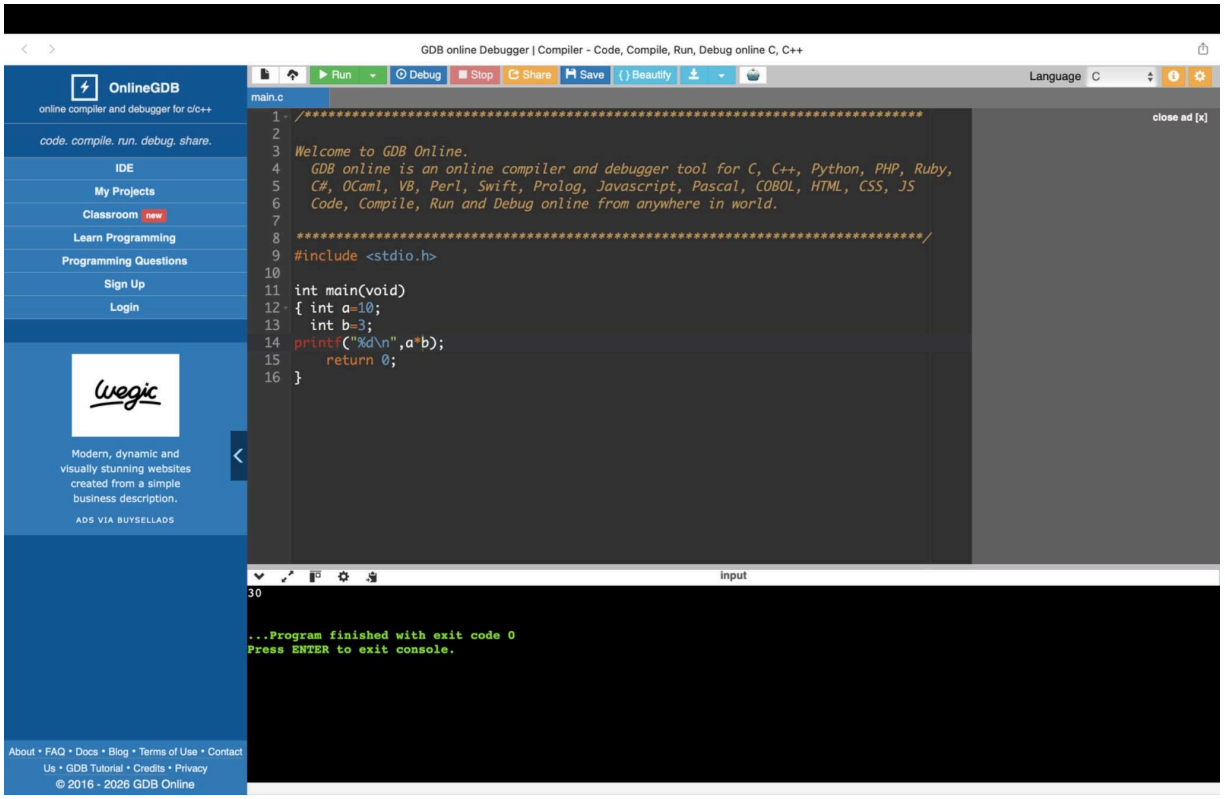
• Acrescente resto da divisão.

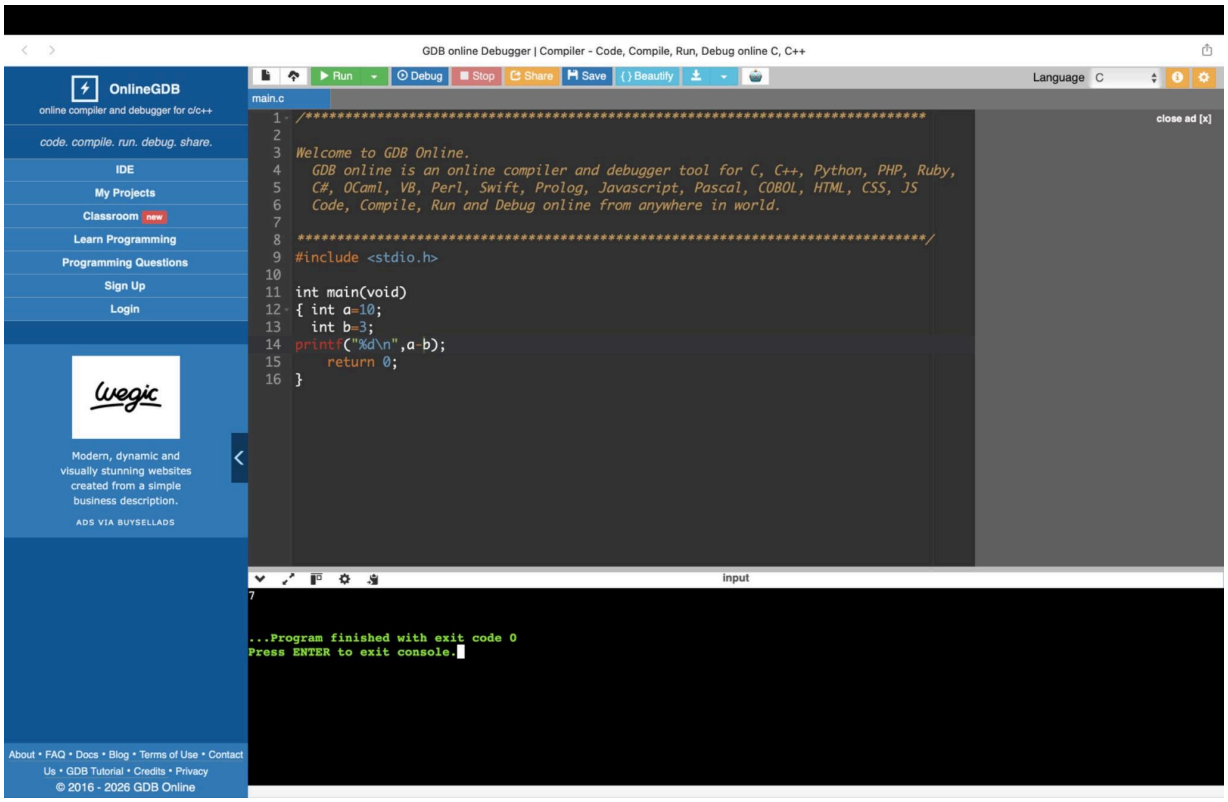
Entrega da parte

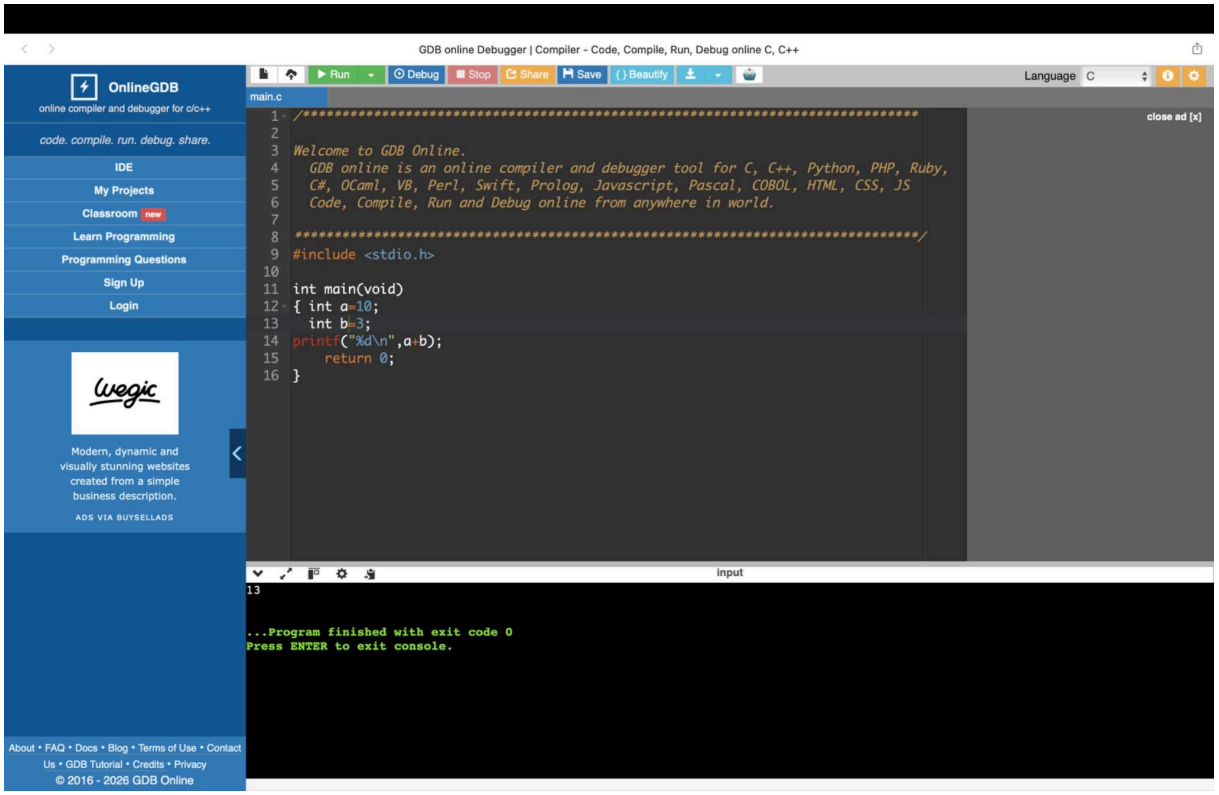
• Tabela de previsões preenchida.
• Programa ampliado com as cinco operações.
• Teste com pelo menos dois pares de números.

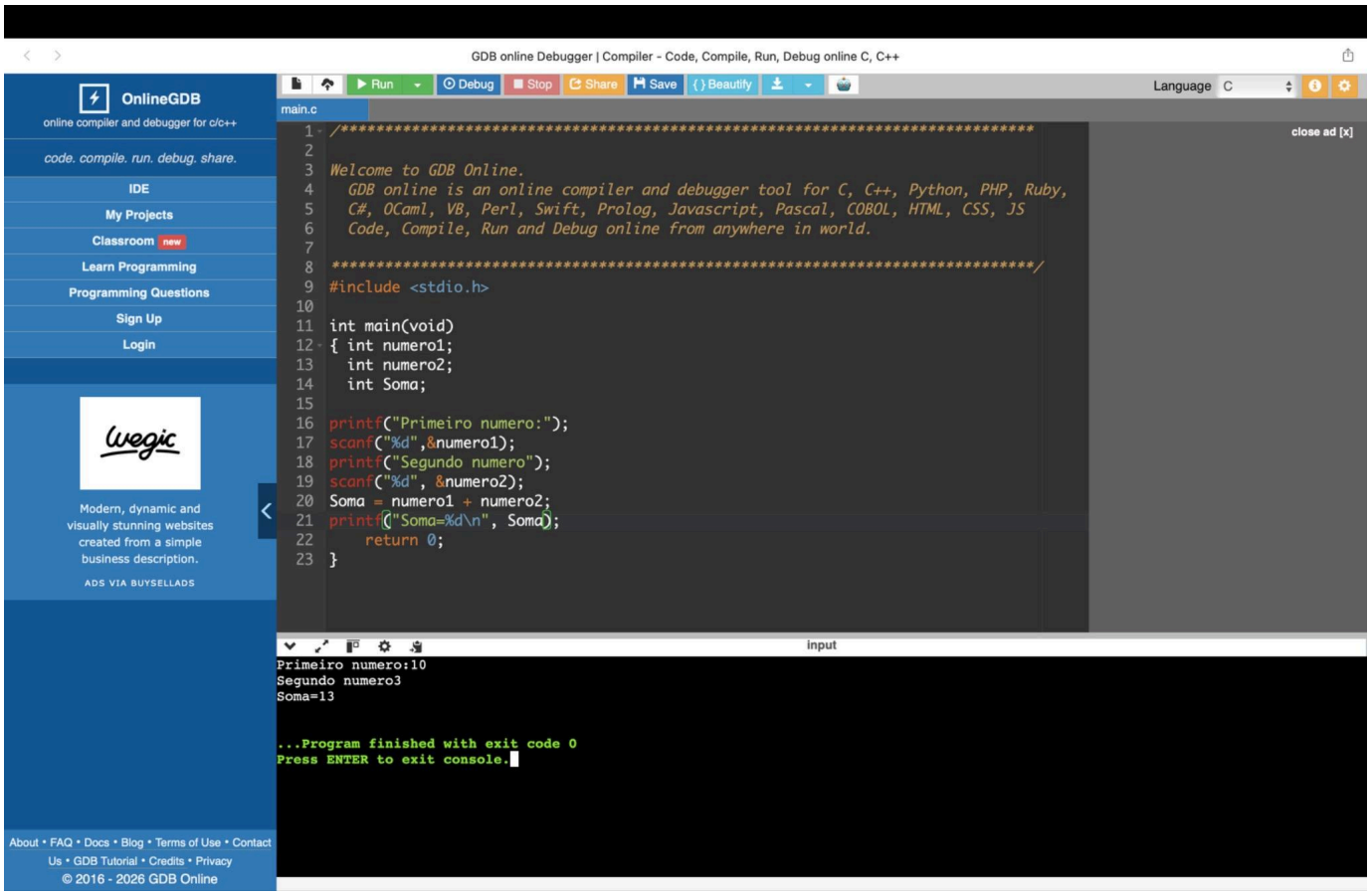


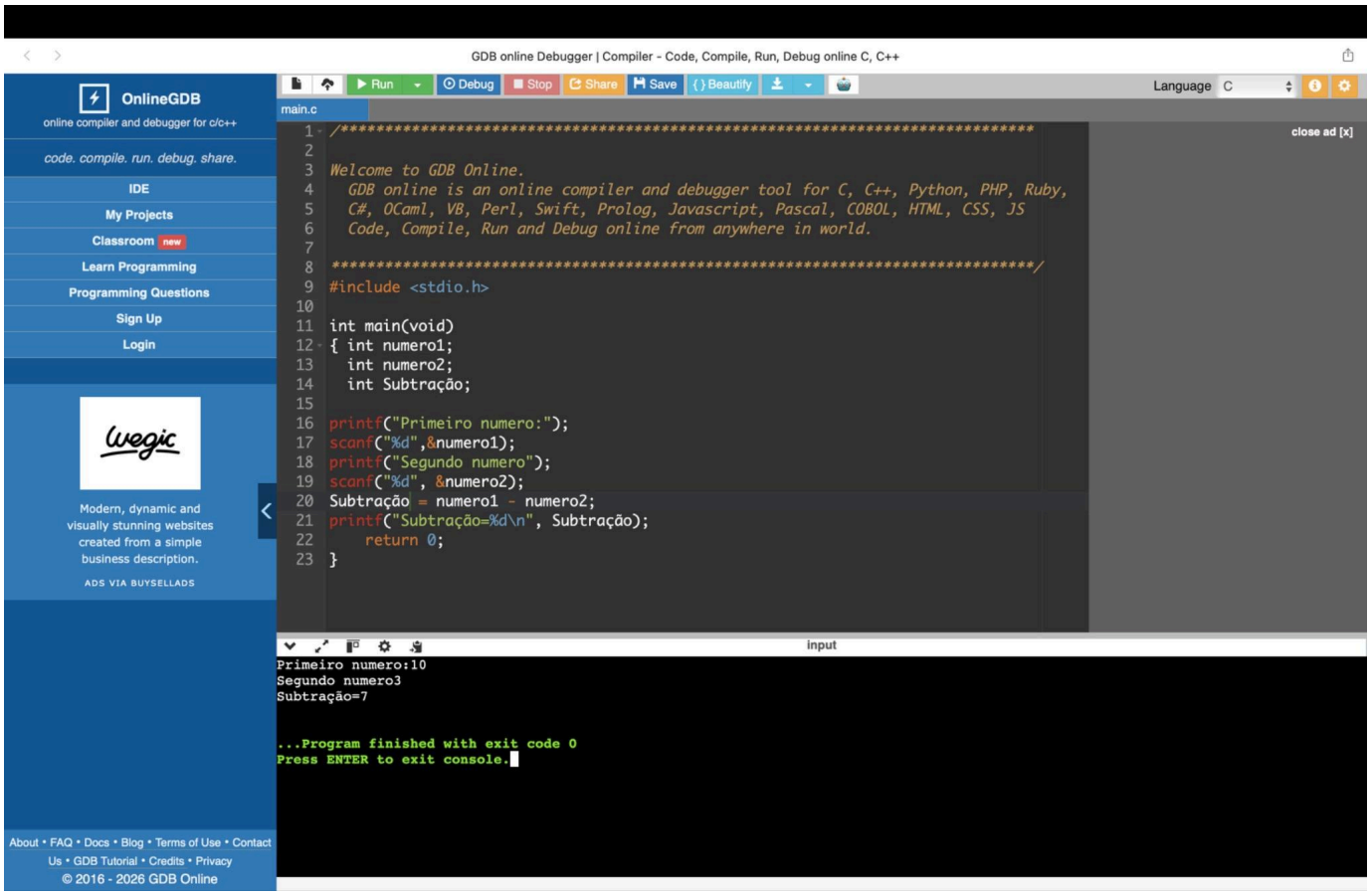

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***/
#include <stdio.h>
int main(void)
{ int a=10;
int b=3;
printf("%d\n",a/b);
return 0;
}
input
3
...Program finished with exit code 0
Press ENTER to exit console.


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***/
#include <stdio.h>
int main(void)
{ int a=10;
int b=3;
printf("%d\n",a*b);
return 0;
}
input
30
...Program finished with exit code 0
Press ENTER to exit console.


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***/
#include <stdio.h>
int main(void)
{ int a=10;
int b=3;
printf("%d\n",a-b);
return 0;
}
input
7
...Program finished with exit code 0
Press ENTER to exit console.


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***/
#include <stdio.h>
int main(void)
{ int a=10;
int b=3;
printf("%d\n",a+b);
return 0;
}
input
13
...Program finished with exit code 0
Press ENTER to exit console.


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***/
#include <stdio.h>
int main(void)
{ int numero1;
int numero2;
int Soma;
printf("Primeiro numero:");
scanf("%d",&numero1);
printf("Segundo numero");
scanf("%d", &numero2);
Soma = numero1 + numero2;
printf("Soma=%d\n", Soma);
return 0;
}
input
Primeiro numero:10
Segundo numero3
Soma=13
...Program finished with exit code 0
Press ENTER to exit console.


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***/
#include <stdio.h>
int main(void)
{ int numero1;
int numero2;
int Subtração;
printf("Primeiro numero:");
scanf("%d",&numero1);
printf("Segundo numero");
scanf("%d", &numero2);
Subtração = numero1 - numero2;
printf("Subtração=%d\n", Subtração);
return 0;
}
input
Primeiro numero:10
Segundo numero3
Subtração=7
...Program finished with exit code 0
Press ENTER to exit console.